12TH CHEMISTRY · CHAPTER 26 · SOLVED EXERCISE

Polymers

Complete textbook exercise (MCQs, Short Answer, Constructed Response, Descriptive) with Quick Check solutions

Q1. MULTIPLE CHOICE QUESTIONS

I. Which pair of monomers reacts to form a polyester?

(a) Diamine and dicarboxylic acid
(b) Diol and dicarboxylic acid
(c) Ethene and chloroethene
(d) Amino-carboxylic acid

Answer: (b) Diol and dicarboxylic acid — The –OH of the diol reacts with the –COOH of the acid to give ester links.

II. What small molecule is eliminated when a diamine reacts with a dioyl chloride?

(a) Water (H_2O)
(b) Hydrogen chloride (Hcl)
(c) Ammonia (NH_3)
(d) Ethene (C_2H_4)

Answer: (b) Hydrogen chloride (Hcl) — $-NH_2$ reacts with –COCl and HCl is removed.

III. How do plasticizers affect PVC?

(a) Make it harder.
(b) Make it brittle
(c) Reduce inter-chain forces, make it flexible
(d) Increase melting point and conductivity

Answer: (c) Reduce inter-chain forces, make it flexible — They reduce the strong forces between the chains so the chains move more freely.

IV. What are the two monomers for Nylon 6,6?

(a) Vinyl chloride & Adipic acid
(b) Hexamethylenediamine & Adipic acid
(c) Ethane-1,2-diol & Adipic acid
(d) Only Hexamethylenediamine

Answer: (b) Hexamethylenediamine & Adipic acid — A six-carbon diamine and a six-carbon dicarboxylic acid.

V. A polymer section is shown: $[-CH_2-CH(CH_3)-CH_2-CH(CH_3)-]$... What is the monomer, and what type of polymer is it?

(a) Propene; Addition
(b) Propene; Condensation
(c) Ethene; Addition
(d) Ethene; Condensation

Answer: (a) Propene; Addition — The chain has only carbon atoms; the repeat unit $–CH_2–CH(CH_3)–$ comes from propene.

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VI. Polyesters and polyamides are considered biodegradable because they:

(a) are natural polymers. (b) break down in sunlight.
(c) have small monomers. (d) are hydrolysable ester/amide links.

Answer: (d) are hydrolysable ester/amide links. — Ester and amide links can be broken by hydrolysis.

VII. What are the products of alkaline hydrolysis (saponification) of a polyester?

(a) Carboxylic acid and alcohol (b) Carboxylate salt and alcohol
(c) Carboxylic acid and amine (d) Carboxylate salt and amine

Answer: (b) Carboxylate salt and alcohol — In alkali the acid is obtained as its salt; the ester link also gives the alcohol.

VIII. What are the products of acidic hydrolysis of a polyamide?

(a) Carboxylate salt and amine
(b) Carboxylic acid and alcohol
(c) Carboxylic acid and protonated amine
(d) Carboxylate salt and protonated amine

Answer: (c) Carboxylic acid and protonated amine — In acid the amine formed is protonated.

IX. What is the primary cause of photodegradation in polymers?

(a) Acidic hydrolysis (b) Microorganisms
(c) Absorption of UV light (d) Alkaline hydrolysis

Answer: (c) Absorption of UV light — UV light absorbed (e.g. by C=O groups) weakens the bonds in the chain.

X. Poly(alkenes) are considered "chemically inert" because they:

(a) have weak, easily broken bonds. (b) have strong C-C and C-H bonds.
(c) They react quickly with acids. (d) They dissolve in water.

Answer: (b) have strong C-C and C-H bonds. — Strong, non-polar C–C and C–H bonds are not attacked by acids, alkalis or water.

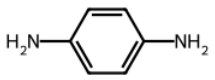
Q2. SHORT ANSWER QUESTIONS

a) Kevlar® is a thermally and mechanically very stable but flexible polymer; a section of this polymer is given below:
–[CO–C_6H_4–CO–NH–C_6H_4–NH]–n
i) Name the characteristic linkage and type of polymerisation in its structure ii) Identify the two functional groups that react to form it. iii) Name the monomers from which it is formed.

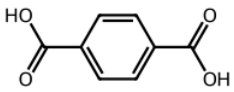
i) Amide (peptide) linkage, –CO–NH–; it is formed by condensation polymerisation.

ii) The amino group (–NH_2) and the carboxylic acid group (–COOH) [or the acyl chloride group, –COCl].

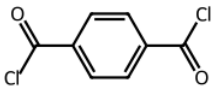
iii) Benzene-1,4-diamine and benzene-1,4-dicarboxylic acid (or benzene-1,4-dicarbonyl chloride).



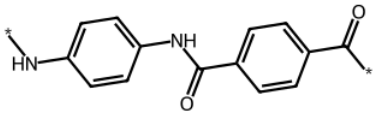
benzene-1,4-diamine



benzene-1,4-dicarboxylic acid



benzene-1,4-dicarbonyl chloride



repeat unit of Kevlar

b) Explain why the reaction between an acyl chloride and ammonia is classified as a condensation reaction. Identify the small molecule that is eliminated during this reaction.

The two molecules join together through a new C–N (amide) bond and, at the same time, a small molecule is eliminated. Joining with loss of a small molecule is a condensation reaction. The small molecule eliminated is **hydrogen chloride (HCl)**.

$$CH_3–COCl + NH_3 \rightarrow CH_3–CO–NH_2 + HCl$$

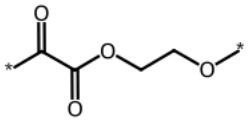
c) For the following monomers: [ethanedioyl chloride, ClOC–COCl, and ethane-1,2-diol, HO–CH_2CH_2–OH]
I) Predict the type of the polymer formed. ii) Draw the repeat unit and a section of the expected polymer. iii) Name this polymer systematically.

In the textbook the structures of these two monomers are printed lower down the page, after part (d).

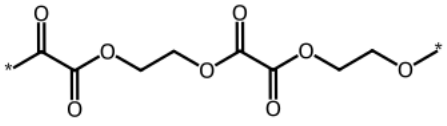
I) A condensation polymer: a **polyester** (–COCl reacts with –OH, HCl is eliminated).

ii) Repeat unit and section:

$$–[O–CH_2–CH_2–O–CO–CO]–n$$



repeat unit



section of the polymer (two repeat units)

(shows the bond to the next unit)*

iii) Poly(ethane-1,2-diyl ethanedioate).

d) For Nylon 6,6 (from 1,6-hexanediamine and 1,4-hexandioic acid acid): i) Name the linkage. ii) Name the small molecule eliminated. iii) Draw the repeating unit.

i) Amide (peptide) linkage, –CO–NH–.

ii) Water, H_2O.

iii) Repeating unit:

$$-[HN-(CH_2)_6-NH-CO-(CH_2)_4-CO]-n$$

The textbook names the acid "1,4-hexanedioic acid"; the six-carbon diacid drawn in its figure, HOOC–$(CH_2)_4$–COOH, is hexanedioic acid (adipic acid).

e) For poly(chloroethene) (PVC) i) Name the monomer for PVC ii) Why can it undergo addition polymerization?

i) Chloroethene (vinyl chloride), $CH_2=CHCl$.

ii) Because it has a carbon–carbon double bond (C=C). This bond opens up and allows the monomer molecules to link together one by one, without loss of any atoms.

f) State two properties of nylon that make it suitable for ropes and fishing nets.

(1) High tensile strength: it is strong and resistant to breaking, rubbing and wear and tear. (2) Resistance to water wear (it is also stretchable due to its elasticity).

g) Give two industrial applications for PVC and two for Nylon.

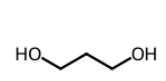
PVC: (1) pipes and fittings for water supply, drainage and sewage systems; (2) electrical insulation for wires and cables.

Nylon: (1) mechanical parts of the automotive industry such as gears and bearings; (2) ropes and fishing nets (also clothing and carpets).

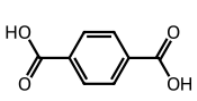
h) Draw the repeating unit of the polyester from propan-1,3-diol and benzene-1,4-dicarboxylic acid.

An ester link forms at each end with loss of water:

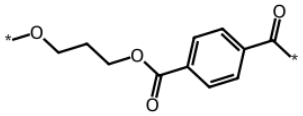
$$-[O-CH_2-CH_2-CH_2-O-CO-C_6H_4-CO]-n$$



propane–1,3–diol



benzene–1,4–dicarboxylic acid



repeat unit of the polyester

(shows the bond to the next unit)*

i) State three distinct environmental problems caused by the disposal of non-degradable polymers.

- **Persistent pollution:** they stay in the environment for a very long time, so they accumulate in landfills, oceans and natural landscapes.

- **Landfill space:** they take up huge amounts of space in landfills because they do not decompose; as population grows, finding space for more landfills becomes difficult.
- **Harm to animals:** animals (especially marine life) can mistake plastic waste for food, leading to choking, internal injuries or starvation; they can also get tangled in plastic debris.

j) Name two different polymers used in artificial organs and state the specific application for each.

(1) **Poly(tetrafluoroethylene), PTFE** – artificial blood vessels (grafts); its smooth surface prevents blood clots. (2) **Poly(methyl methacrylate), PMMA** – artificial cornea; it is perfectly clear and biocompatible.

Q3. CONSTRUCTED RESPONSE QUESTIONS

a) Describe the chemical role of plasticizers in polyvinyl chloride (PVC) and explain how this modification influences the polymer's industrial applications.

PVC is a versatile plastic whose properties can be changed a lot by adding different chemicals (plasticizers). **Plasticizers** are chemicals added to materials to make them softer, more flexible and easier to work with. They do this by reducing the strong forces between the polymer chains, allowing the chains to move more freely.

Influence on applications: rigid PVC (without plasticizers) is hard and stiff, so it is used for pipes and fittings, window frames and door profiles. Plasticised PVC is soft and flexible and can be bent and shaped easily, so it is used for insulation of wires and cables, tubing, blood bags, cling films, raincoats, artificial leather and inflatable products.

b) Explain why poly(alkenes) like poly(ethene) pose a greater environmental challenge than polyesters and polyamides. Identify the chemical process that allows the latter to degrade.

Poly(alkenes) have only strong, non-polar C–C and C–H bonds in their chains and no ester or amide links. They are therefore chemically inert: they are not attacked by acids, alkalis or water (no hydrolysis), and soil organisms cannot break them down. This is why they are difficult to biodegrade.

So poly(alkene) waste stays in the environment for a very long time, fills landfills, harms animals and forms microplastics. Polyesters and polyamides contain ester and amide links which soil organisms can attack, so they break down into simpler components.

Chemical process: hydrolysis (acidic or alkaline) of the ester or amide links.

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DESCRIPTIVE QUESTIONS

Q.4 Compare and contrast addition and condensation polymerization, giving details of the specific structural requirements for the monomers in each.

Addition polymerisation	Condensation polymerisation
Monomers add to each other one by one; all atoms of the monomers become part of the polymer.	Each time a new bond forms, a small molecule is eliminated from the monomers.
Monomers must have a C=C double (or C≡C triple) bond.	Monomers must have at least two reactive functional groups (–OH, –COOH, $–NH_2$, –COCl) on different ends.
No small molecule is produced as a side product.	A small molecule (H_2O, HCl, CH_3OH or NH_3) is always produced.
Often a chain reaction started by an initiator (free radicals).	Typically takes place in a step-by-step manner.
The main chain consists only of carbon atoms.	The main chain contains O or N atoms with carbon (ester or amide links).
Examples: poly(ethene), PVC, poly(propene), PTFE.	Examples: polyesters (Terylene/PET) and polyamides (nylon-6,6, Kevlar).

Similarity: in both, many small monomer molecules join to form a very large polymer molecule.

Structural requirement for addition: the monomer must contain a C=C double (or C≡C triple) bond, which opens up to link the monomers.

$$n\ CH_2{=}CH_2 \rightarrow –[CH_2–CH_2]–n$$

Structural requirement for condensation: the monomers must have at least two reactive functional groups on different ends, e.g. a diol with a dicarboxylic acid, a diamine with a dicarboxylic acid or dioyl chloride, or one monomer carrying both groups (an amino acid or a hydroxy acid).

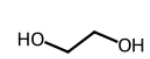
$$n\ HO–CH_2CH_2–OH + n\ HOOC–C_6H_4–COOH \rightarrow$$
$$–[O–CH_2CH_2–O–CO–C_6H_4–CO]–n + 2n\ H_2O$$

Q.5 Illustrate the formation of two major condensation polymers.

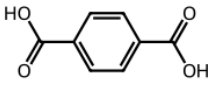
1. A polyester: Terylene (PET)

Polyesters have **ester links** (–CO–O–) in their backbone. A diol (a molecule with two –OH groups) and a dicarboxylic acid (a molecule with two –COOH groups) are combined to give a polyester. The –OH of the diol reacts with the –COOH of the acid and a molecule of water is removed. Example: Terylene (PET) from ethane-1,2-diol and benzene-1,4-dicarboxylic acid.

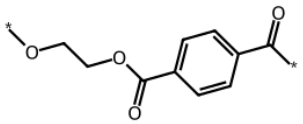
$$n\ HO–CH_2CH_2–OH + n\ HOOC–C_6H_4–COOH \rightarrow$$
$$–[O–CH_2CH_2–O–CO–C_6H_4–CO]–n + 2n\ H_2O$$



ethane-1,2-diol



benzene-1,4-dicarboxylic acid



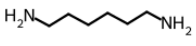
repeat unit of PET (Terylene)

Monomers and repeat unit of PET (shows the bond to the next unit)*

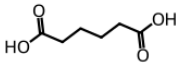
2. A polyamide: nylon-6,6

Nylon-6,6 is the most common example of synthetic amides. It is formed by a polymerisation reaction between 1,6-diaminohexane and hexanedioic acid (adipic acid); each monomer has six carbon atoms.

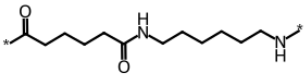
$$n\ H_2N–(CH_2)_6–NH_2 + n\ HOOC–(CH_2)_4–COOH \rightarrow$$
$$–[HN–(CH_2)_6–NH–CO–(CH_2)_4–CO]–n + 2n\ H_2O$$



1,6-diaminohexane



hexanedioic acid (adipic acid)



repeat unit of nylon-6,6

Monomers and repeat unit of nylon-6,6 (shows the bond to the next unit)*

The textbook names the acid "1,4-hexanedioic acid"; the six-carbon diacid drawn in its figure, $HOOC–(CH_2)_4–COOH$, is hexanedioic acid (adipic acid).

Q.6 Differentiate between the environmental degradation of poly(alkenes) and polyesters.

Poly(alkenes)	Polyesters
The chain has only strong, non-polar C–C and C–H bonds.	The chain contains ester links (–COO–).
Chemically inert; they cannot be hydrolysed by acids or alkalis.	The ester links are hydrolysed by acids and by alkalis.
Soil organisms cannot attack them, so they are non-biodegradable.	Soil organisms attack them through hydrolysis, so they are biodegradable.
They stay in the environment for a very long time and slowly break into microplastics.	They are decomposed into carboxylic acids and alcohols (simpler components).
They fill landfills, pollute oceans and harm animals.	They do not stay in the environment forever; e.g. PLA plastics and dissolvable stitches.

QUICK CHECK QUESTIONS (IN-CHAPTER)

These questions appear in the "Quick Check" boxes inside the chapter. The textbook numbers two different boxes as Quick Check 26.2; both are solved, in the printed order.

Quick Check 26.1

a) The repeating unit of nylon-6, 10 is shown below. Draw the structures of its monomers from this structure.
–[NH–$(CH_2)_6$–NH–CO–$(CH_2)_8$–CO]–

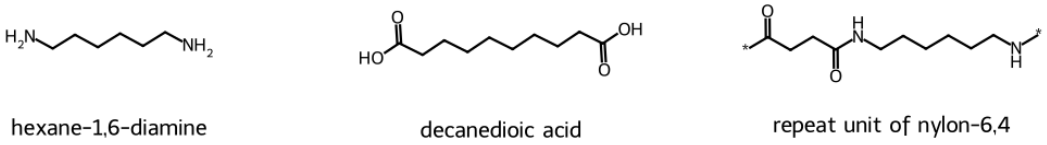
Break the amide bonds; add –H to each N and –OH to each C=O. The monomers are **hexane-1,6-diamine** (6 carbon atoms) and **decanedioic acid** (10 carbon atoms).

$H_2N–(CH_2)_6–NH_2$ and $HOOC–(CH_2)_8–COOH$

b) Monomers of nylon-6,4 are given below. Draw the structure of the repeating unit of the polymer formed from these monomers.
$H_2N(CH_2)_6NH_2$ $HOOC(CH_2)_2COOH$

An amide link forms between each amino group and each carboxylic group, with loss of water. The repeating unit is:

–[HN–$(CH_2)_6$–NH–CO–$(CH_2)_2$–CO]–n



hexane-1,6-diamine decanedioic acid repeat unit of nylon-6,4

Monomers of nylon-6,10 and the repeat unit of nylon-6,4 (shows the bond to the next unit)*

Quick Check 26.2

a) Identify the type of polymerisation and monomers forming the following polymer sections.
i) –CH_2–NH–CO–CH_2–NH–CO–CH_2– ii) a carbon chain with a benzene ring on every second carbon iii) –$CH_2CH(CH_3)$–$CH_2CH(CH_3)$–$CH_2CH(CH_3)$– (polypropylene)

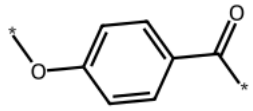
i) The chain contains amide (–CO–NH–) links, so it is a polyamide formed by **condensation** polymerisation. Monomer: aminoethanoic acid (glycine), $H_2N–CH_2–COOH$.

ii) The main chain has only carbon atoms, so it is an **addition** polymer (poly(phenylethene), polystyrene). Monomer: phenylethene (styrene), $C_6H_5–CH=CH_2$.

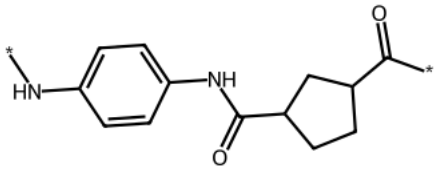
iii) The main chain has only carbon atoms, so it is an **addition** polymer. Monomer: propene, $CH_3–CH=CH_2$.

Quick Check 26.2 (second box)

Give the structures of the polymers formed from the monomers. For each polymer identify the following: i. the repeat unit ii. the type of linkage present iii. the attractive force between the polymer chains.
a) HO–C_6H_4–COOH (4-hydroxybenzoic acid) b) H_2N–C_6H_4–NH_2 + cyclopentane-1,3-dicarboxylic acid



(a) repeat unit: polyester



(b) repeat unit: polyamide

Repeat units (shows the bond to the next unit)*

	Polymer (a)	Polymer (b)
i. Repeat unit	–[O–C_6H_4–CO]–	–[HN–C_6H_4–NH–CO–C_5H_8–CO]–
ii. Linkage	Ester linkage (–CO–O–); a polyester	Amide linkage (–CO–NH–); a polyamide
iii. Force between chains	Permanent dipole–dipole forces between the polar ester groups (with instantaneous dipole forces)	Hydrogen bonding between the N–H of one chain and the C=O of another

The forces between polymer chains are not discussed in this chapter; part iii is answered from the intermolecular forces studied earlier.

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